

Analysis of LDPC Codes in FSO Communication System Under Fast Fading Channel Conditions

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ABSTRACT The free-space optical (FSO) communication represents a promising futuristic technology due to several benefits, which has attracted significant attention recently for various applications. FSO communication is susceptible to fading due to atmospheric turbulence. FSO channel, being a slow fading channel, inherently does not allow the efficient use of error-correcting codes (ECCs). ECCs have shown to have good performance in a fast fading channel. A way to convert the slow fading channel into a fast fading channel is by using techniques like diversity. Diversity makes the bits pass through many channels simultaneously, and thus the fast fading channel effect is observed at the receiver. In this paper, we first derive the capacity and bit error rate performance of the low-density parity-check (LDPC) codes in a fast fading channel. We then study the performance of LDPC codes in receiver diversity systems. Finally, we see the impact of using LDPC coded multiple-input multiple-output (MIMO) systems in the FSO channel. We use regular (3,6) LDPC code and irregular LDPC codes of rate 0.5, where the desired degree distribution is optimized for the channel condition.

INDEX TERMS LDPC code, fast fading channel, free-space optical communication, spatial multiplexing.

I. INTRODUCTION

FREE-SPACE optical (FSO) communication aids high bandwidth (≈ 200 THz), secure transmission link due to its operational wavelength range of 700 – 1500 nm [1], and light and small-sized equipment. The other advantages of FSO communication are: no road digging required for the transmission [1], [2], easy deployment, unregulated spectrum, etc. These benefits engage several applications such as last-mile access, disaster recovery, intercampus connections, high definition television signals, or hilly terrain installations.

FSO communication, however, is prone to deteriorating effects of atmospheric turbulence [3], [4]. The atmospheric turbulence causes fluctuations in the received optical beam, inevitably degrading the quality of the signal. This effect is often termed signal fading, and several mathematical models have been proposed in the literature for its analysis. We use Gamma-gamma (GG) probability density function (PDF) to model the effect of fading in our work [5]. Additionally, the received signal is influenced by various optical noises such

as background noise and receiver noise, including thermal noise, shot noise, dark current noise, etc.

One promising strategy to curtail the effects of fluctuations in the received signal is to use error-correcting codes (ECCs). Low-density parity-check (LDPC) code is a primary example of ECCs, which has become very popular as it attains near Shannon capacity (within 0.0045 dB) in the additive white Gaussian noise (AWGN) channel [6]. The authors in [7] analyzed the belief propagation (BP) algorithm with Gaussian approximation. However, this approximation cannot be applied to an FSO channel. In [8], we report that the channel log-likelihood ratio (LLR) for the FSO channel is not symmetric and hence not Gaussian distributed.

We have designed LDPC codes for FSO channels in [9]. We have also formulated and presented the analysis of LDPC codes based on different channel conditions in [10]. In [11] and [12], the authors have analyzed the performance of LDPC coded multiple-input multiple-output (MIMO) FSO system through simulations.

In [10], we show that the FSO channel is considered a slow fading channel, as the channel's coherence time is in ms [1], and is much larger than the usual bit duration (which is 1 ns). This channel condition is not ideal for ECCs as the channel gain remains constant for almost 1 Mb information transmission (for a data rate of 1 Gb/s). Thus, channel coding cannot improve a slow fading channel's performance as long as the probability of the channel in deep fade is non-zero [13]. One method is to use an interleaver. However, due to long lengths of fading (typically more than 1 ms) and a high data rate (pulse widths less than 1 ns), an interleaver with too long depths (greater than 100 Mb) is necessitated entailing long memories and packet delays. Thus, interleaver is not very helpful in building robust communications systems. The method that we propose uses spatial multiplexing systems in conjunction with LDPC codes. As we show in this paper, the proposed method also achieves a considerable coding gain and has no implications for delays. Thus, this method is more likely to be used in FSO communications systems, mainly where packet delays are also significant. We compare the performance of the coded spatial multiplexed FSO channel with the fast fading channel as it is considered optimum channel condition for the application of ECCs [10].

A. RELATED WORK

This section presents a literature review related to our work. The authors in [11] and [14] study the performance of the structured LDPC-coded MIMO FSO system under the high turbulence regime. The modulation scheme considered is Q -ary pulse amplitude modulation (PAM) and Q -ary pulse position modulation (PPM) scheme, for better spectral and power efficiency, respectively. With M transmitters and K receivers, the MIMO configurations considered [11], [14] are: $M = 2, K = 4$, $M = 2, K = 2$, $M = 4, K = 1$, and $M = 2, K = 1$, and the code rate is 0.75. The performance of our system is better when we compare it with the performance of 2-PAM [11] and 2-PPM [14] LDPC coded system with $M = 1$ and $K = 1$ configuration. We only compare the LDPC coded system without diversity as the number of transmitters and receivers considered in the related works do not match our diversity configurations ($M = 2, K = 5$ and $M = 1, K = 10$).

We consider pseudo-random LDPC code construction in our work. The newer developments in LDPC code are to go toward structured LDPC codes. Structured LDPC codes though reduce complexity but deteriorate BER performance [15]. The other direction is to focus on integrating LDPC with techniques like orbital angular momentum [16], using coherent receivers based on Kramers–Kronig [17], or using a combination of LDPC code with trellis-coded modulation [18], etc. These directions are promising but usually focus on a single link and cannot accrue advantages of diversity in deep fade conditions.

B. CONTRIBUTIONS

Our work aims to study and analyze a degree-optimized irregular LDPC coded diversity system, which realizes

advantages similar to the fast fading FSO channel. We propose a thorough analytical model for the above, which any other research paper has not proposed to the best of our knowledge. We summarize the novel contributions of the paper below.

- i. Analysis of the BP decoding algorithm of degree-optimized irregular LDPC codes in a spatially multiplexed and fast fading FSO channel.
- ii. Analytical results under the above two mentioned channel conditions are compared with the simulation models.
- iii. Derivation of the Shannon capacity for both the channel conditions. We also evaluate gap-to-capacity, i.e., how far we are operating from the Shannon limit.

In the analysis, we assume that the channel state information (CSI) is available at the receiver. It is expected that such a comprehensive study of the LDPC codes will benefit researchers and industries alike. To the best of our knowledge, a detailed analysis of LDPC codes in fast fading, receiver diversity, and MIMO channels has not been done earlier.

The remainder of this paper is organized as follows. We describe the statistical system and channel models in Section II and enumerate the capacity for a fast fading channel and spatial multiplexed FSO channel in Section III. We discuss the basics of LDPC codes and formulate the effect of LDPC codes in a slow fading channel in Section IV. We examine the performance of LDPC codes in a fast fading and spatial multiplexed channel in Section V. Further, analytical results presented in this section are supported with the simulations. Finally, we conclude the paper in Section VI.

II. STATISTICAL SYSTEM AND CHANNEL MODELS

In this section, we discuss the system and channel models which we consider for our study.

A. SYSTEM MODEL

We show an end-to-end FSO communication system in Fig. 1. A laser is used for intensity modulation (IM) at the transmitter, and a direct detection (DD) receiver with PIN photo-detector is employed. The interleaver is used in only one section of our work, and hence it is shown with the dotted lines. The received symbols are deinterleaved and passed on to the LDPC decoder.

In an FSO channel, the bit b is mapped to the intensity-modulated symbol X , where $X = (1 - b)2\Re P\sqrt{T}$, $\Re$ is the responsivity of the photo-detector, P the average received optical power, and T is the duration of the transmitted orthonormal rectangular pulse.

$$Y = HX + I_{th}, \quad (1)$$

where Y is the received data vector, H the intensity fading vector, and I_{th} represents the thermal noise current vector, which is AWGN with $\overline{i_{th}^2} = 4KT_eB/R_L$ where $\overline{i_{th}^2}$ is the mean squared thermal noise variance [19], K the Boltzmann's

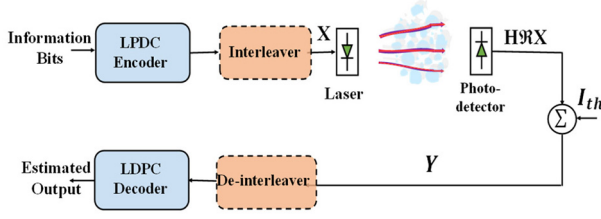


FIGURE 1. Block diagram of the FSO communication system.

constant, T_e the room temperature in degree Kelvin, and R_L the load resistor.

B. CHANNEL MODEL

The channel is characterized by its PDF and coherence time. The irradiance fluctuations (real-valued) caused by the atmospheric turbulence are modeled as GG distribution. The corresponding PDF [5] is given as

$$f_H(h) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} h^{((\alpha+\beta)/2)-1} K_{\alpha-\beta}(2\sqrt{\alpha\beta}h), \quad (2)$$

where the symbol Γ represents the gamma function and $K_{\alpha-\beta}$ the modified Bessel function of the second kind and order $(\alpha - \beta)$. The parameters α and β are directly related to atmospheric conditions and determine the turbulence strength. α and β are the pdf parameters describing the scintillation experienced by plane waves, and in the case of zero-inner scale, they can be derived from Rytov variance and are given as

$$\alpha = \left[\exp \left(\frac{0.49\sigma_1^2}{(1 + 1.11\sigma_1^{12/5})^{7/6}} \right) - 1 \right]^{-1}, \quad (3)$$

$$\beta = \left[\exp \left(\frac{0.51\sigma_1^2}{(1 + 0.69\sigma_1^{12/5})^{5/6}} \right) - 1 \right]^{-1}, \quad (4)$$

where σ_1^2 represents the Rytov variance and is given by

$$\sigma_1^2 = 1.23C_n^2 k^{7/6} L^{11/6}. \quad (5)$$

In the above equation, $k = 2\pi/\lambda$, where λ is the wavelength, L the propagation length and C_n^2 the refractive index structure parameter. We consider a propagation length of 2.2 km in this work. The parameter C_n^2 depends on temperature, humidity, and other weather conditions, and its value varies with the time of day and height at which the signal is transmitted. We report the values of C_n^2 at a $\lambda = 1550$ nm in Table 1. This table also contains the GG distribution parameters used to model high to low turbulence conditions.

C. COHERENCE TIME

The coherence time of the channel is an important characteristic. It decides the bits' time length up to which the transmitted bits can be considered to experience the same

TABLE 1. G-G distribution parameters.

Turbulence Regime	α	β	σ_1^2	C_n^2
High	4.2	1.4	3.5	4.1×10^{-7}
Moderate	4.0	1.9	1.6	1.9×10^{-7}
Low	11.6	10.1	0.3	3.5×10^{-8}

channel condition. The channel coefficients are slowly-faded gamma-gamma distributed in a slow fading channel, with a coherence time longer than 1 ms [13]. Consequently, for a data rate of 1 Gb/s over this channel, a single channel state encapsulates a block of 10^6 bits, which is typically longer than the blocks used in LDPC. Thus, the LDPC codes attain a performance similar to that of an uncoded system [10]. Alternatively, the use of interleavers over a slow-faded FSO channel makes a single channel state experience one data bit. This allows the coherence time to be approximately equal to the data bit duration. Also, in a fast fading channel condition, coherence time is smaller than the bit duration, allowing each data bit to experience multiple channel states. Hence, channel averaging makes the channel performance much better than the slow fading channel [13].

In [9], we present the method to model the channel with the desired coherence time, and we generate the GG process as

$$z_k = x_k \times y_k, \quad \forall k. \quad (6)$$

where x_k and y_k gives the samples of gamma process based on the shape and scale parameter, respectively. $x_{k+1} = \frac{1}{(t_c+t_b)} \times (x_k t_c + t_b + \frac{t_b}{2\alpha}(\zeta^2 - 1)) + (\sqrt{2x_k \frac{t_c t_b}{\alpha}}) \times \zeta$, $\forall k$, and $y_{k+1} = \frac{1}{(t_c+t_b)} \times (y_k t_c + t_b + \frac{t_b}{2\beta}(\zeta^2 - 1)) + (\sqrt{2y_k \frac{t_c t_b}{\beta}}) \times \zeta$, $\forall k$, and indices k and $k+1$ refer to the time intervals t_k and t_{k+1} , respectively, t_c the coherence time of the channel, t_b the bit duration, and ζ the pseudorandom value drawn from the standard Gamma distribution.

III. CAPACITY CALCULATION

This section calculates the capacity for a fast fading channel and spatially multiplexed channel.

A. FAST FADING CHANNEL

In a fast fading channel, the duration of information bit extends through multiple channel states, and thus, it experiences different fading process realizations. By the law of large numbers, time averages (in interleavers) equal the ensemble averages (in spatial multiplexing), a concept called ergodicity. Thus, the ergodic capacity with the perfect CSI at the receiver for a given SNR is derived using

$$C = \lim_{N \rightarrow \infty} B \frac{1}{N} \sum_{n=0}^{N-1} (1 + |h_n|^2 \text{SNR}) = E[B(1 + |h|^2 \text{SNR})], \quad (7)$$

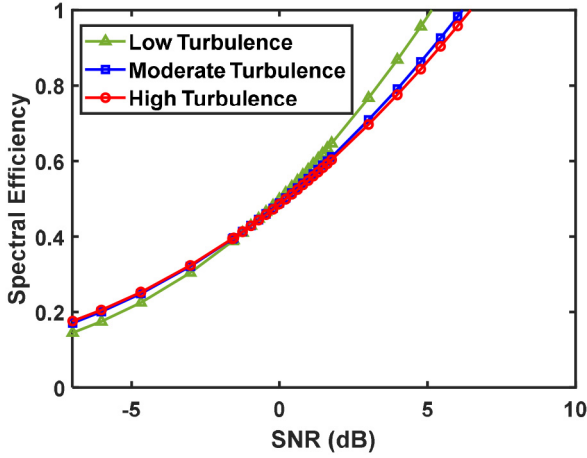


FIGURE 2. Spectral efficiency vs. SNR in high, moderate, and low turbulence regimes.

TABLE 2. Capacity evaluation of fast fading channel.

Turbulence Regime	Spectral Efficiency, ρ , bpcu	Transmitted Optical power, P (μ W)	$\frac{P}{\sigma}$	SNR (dB)
High	0.5	0.83	1.45	0.22
High	1	1.71	2.97	6.46
Moderate	0.5	0.82	1.44	0.17
Moderate	1	1.66	2.88	6.19
Low	0.5	0.81	1.41	0.01
Low	1	1.47	2.55	5.14

For the calculation of SNR, we consider $\mathfrak{R} = 1$ and $T = 1$

where h_n is the n^{th} channel state and $N = \frac{t_c}{t_b}$. If $N \rightarrow \infty$, the variation of spectral efficiency, ρ , bits per channel use (bpcu) with SNR is shown in Fig. 2. At low SNR, the capacity of the fading channel is $E[\log_2(1 + h^2 SNR)] \approx E[h^2 SNR] \log_2 e = E[h^2] SNR \log_2 e$. As $E[h^2]$ is the smallest for low turbulence regime; it has the smallest spectral efficiency. On the other hand, at high SNR, the spectral efficiency is approximately $E[\log_2(h^2 SNR)] = \log_2 SNR + E[\log_2 h^2]$. In the high turbulence regime, we get the smallest $E[\log_2 h^2]$ to be -1.5 , and thus it has the lowest spectral efficiency.

The results in Fig. 2 are quite encouraging. They establish that we need an SNR of only 6.46 dB, which translates to an optical power requirement of approximately 1.71μ W, for a spectral efficiency of 1 even in a high turbulence regime. We also show the factor P/σ , as this is proportional to the normalized (with respect to standard deviation of noise) distance between the symbols, in Table 2 and 3. Thus, reliable communication is possible even in a high turbulence regime with moderate power requirements. For more clarity on the numbers required, we also present the results in Table 2.

B. SPATIAL MULTIPLEXING CHANNEL

MIMO systems exploit multiple lasers (M) and photodiodes (K) at transmitter and receiver sections, respectively, for

TABLE 3. Capacity evaluation of spatial multiplexing.

Turbulence Regime	Spectral Efficiency, ρ , bpcu	Transmitted Optical Power, P (μ W)	$\frac{P}{\sigma}$	SNR (dB)
High	0.5	0.82	1.42	-16.23
High	1	0.85	1.47	-5.04
Moderate	0.5	0.81	1.41	-16.72
Moderate	1	0.84	1.46	-5.45
Low	0.5	0.81	1.41	-16.84
Low	1	0.83	1.45	-5.65

For the calculation of SNR, we consider $\mathfrak{R} = 1$ and $T = 1$

capacity and diversity gain. For a MIMO channel where we assume equal power on each receive antenna, the resulting capacity is

$$C = E \left(B \log_2 \left[\det \left(I_K + \frac{SNR}{M} H^T H \right) \right] \right), \quad (8)$$

where I_K is an $K \times K$ identity matrix and $H = M \times K$ channel matrix with random elements h_{mk} . Note that, H^T is the transpose of the matrix H . Also, for the MIMO system, the total transmitted power, and the total receiver aperture area are kept same as that of a single input single output (SISO) system. Moreover, to preserve the incoming and outgoing rate, one bit stretches by the number of transmission links, reducing bandwidth and hence noise power by the same factor. This reduces the required optical power. Similarly, for the single input multiple output (SIMO) configuration, i.e., receiver diversity, the capacity for a given SNR is given as [13]

$$C = E \left(B \log_2 \left(1 + SNR \sum_{k=1}^K |h_k|^2 \right) \right). \quad (9)$$

IV. LDPC CODES

Probabilistic codes such as LDPC codes [20], due to their iterative decoding structures, have become very popular. For a long code length, these codes give near Shannon limit performance [21]. LDPC codes are represented by the sparse parity check matrix H_{par} , characterized graphically using a Tanner graph [22]. It can be viewed as a bipartite graph with two nodes, check nodes (CNs) and bit nodes (BNs). The number of 1s in a row and column vectors, i.e., the node degrees of BNs and CNs of H_{par} , are in general variable. We use $\lambda(x) = \sum_{i=2}^{d_l} \lambda_i x^{i-1}$ and $\rho(x) = \sum_{i=2}^{d_r} \rho_i x^{i-1}$ as the degree distribution functions where λ_i and ρ_i are the fractional degrees of BNs and CNs from an edge perspective and the maximum degree of BN and CN are d_l and d_r , respectively. A special class of LDPC codes is the regular LDPC codes where every BN and CN has the same degree distribution. The rate R of the LDPC code is given by $R = 1 - \frac{\bar{d}_r}{\bar{d}_l}$ where $\bar{d}_l$ and $\bar{d}_r$ give the average degree for the

BNs and CNs, respectively. For decoding, we use the BP algorithm as it is found to be optimal in performance [21].

Parity check matrix (PCM) can be constructed by choosing a pseudo-random or a structured design. The pseudo-random construction is when we avoid small girths in a randomly constructed matrix. For long lengths, the performance of pseudo-random or random PCM remains the same [23]. The typical constraints considered for the random PCM construction are: block length n , row and column weights, cycle length, etc. Random PCM construction achieves the best feasible BER performance [24] at the cost of a relatively high encoding complexity, $O(n^2)$ [25]. Even though structured LDPC does not improve the threshold compared to a pseudo-random construction, but it supports more straightforward and low-cost implementation in hardware and can attain a linear $O(n)$ encoding complexity [26], [27]. In this paper, we focus on the pseudo-random construction of PCM.

A. SLOW FADING CHANNEL

1) BP ALGORITHM IN A SLOW FADING CHANNEL

In the BP algorithm, the messages that are passed to the decoder are the LLR of the received bits. If $x_s \in X$ is the transmitted bit, and the corresponding channel output is y then the LLR is $\mathcal{L} = \log\left(\frac{p(x=0|y)}{p(x=1|y)}\right) = \log\left(\frac{p(y|x=0)}{p(y|x=1)}\right)$, under the constraint that bits '0' and '1' occur with the same probability. We assume that the standard deviation of noise (referred to as σ) affecting both the bits remains the same. The received signal y is Gaussian distributed, and the pdf for bit '0' is given as

$$p(y/0) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(y - x_s h)^2}{2\sigma^2}\right), \quad (10)$$

Similarly, we can find pdf for bit '1'. Now the LLR is given as $\mathcal{L} = \frac{2yhx_s - h^2 x_s^2}{2\sigma^2}$ and the pdf of LLR is

$$N(m_{uo}, 2m_{uo}) = \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{hx_s}\right) \exp\left(-\frac{\left(\mathcal{L} - \frac{(hx_s)^2}{2\sigma^2}\right)^2}{2\frac{(hx_s)^2}{\sigma^2}}\right), \quad (11)$$

where $m_{uo} = \frac{h^2 x_s^2}{2\sigma^2}$. As the entire block sees the same channel state, the pdf in (11) remains valid for the whole of the block. This shows that the LLR is Gaussian distributed, and the channel coefficient alters the mean and the variance of the distribution.

The BP algorithm is an iterative decoding algorithm, where messages flow out between CNs and BNs iteratively, and the output messages are updated as [28]. The mean of the messages from CN and BN after l iterations is termed as $m_v(l)$ and $m_u(l)$, respectively. The mean of the message of BN at the l^{th} iteration is updated as in [28]. The pdf of v_i , where i depicts the output message from the i^{th} edge of the BN in l^{th} iteration is obtained by the convolution of the pdf of LLR (Gaussian distributed) with the symmetric Gaussian pdf of mean $(i-1)m_u(l-1)$ and variance $2(i-1)m_u(l-1)$

$$p(x, m_{v,i}(l), 2m_{v,i}(l))$$

$$= (N(m_{uo}, 2m_{uo}) * N((i-1)m_u(l-1), 2(i-1)m_u(l-1)))(x), \quad (12)$$

where $m_{v,i}(l) = m_{uo} + (i-1)m_u(l-1)$, and $N(m, \sigma^2)$ is Gaussian pdf with mean, m and variance σ^2 , respectively. The final equation of the density evolution is given in (13).

$$m_u(l) = \sum_j^{d_r} \rho_j \phi^{-1} \left(1 - \left[\int_{-\infty}^{\infty} \sum_i^{d_l} \lambda_i \tanh\left(\frac{x_s}{2}\right) \times p(x_s, m_{v,i}(l), 2m_{v,i}(l)) dx \right]^{j-1} \right). \quad (13)$$

The BER given h , that is, $P_{B/h} = Q\left(\sqrt{\frac{m_{uo} + m_u(l)}{2}}\right)$ and this can be marginalized to give overall BER,

$$P_B = \int_0^{\infty} Q\left(\sqrt{\frac{m_{uo} + m_u(l)}{2}}\right) f_H(h) dh. \quad (14)$$

Note that the above equation gives a BER if the length of the code is infinity.

B. SLOW FADING CHANNEL WITH AN INTERLEAVER

With CSI known at the receiver, errors occur whenever a code word is in the deep fade. Indeed, coding across the different coherence periods is now necessary. In a slow fading channel, the consecutive data bits see almost the same channel condition [29]. Thus, the application of ECCs cannot improve the system's performance in a slow fading channel [13]. When we use an interleaver for a slow fading channel, it shuffles the order of the codeword bits before bits are transmitted through the channel [30]. An interleaver makes the bits uncorrelated, and thus channel errors are distributed instead of being localized.

1) BP ALGORITHM IN SLOW FADING CHANNEL WITH AN INTERLEAVER

We study BP algorithm analysis in a slow fading channel with an interleaver in [8]. The final equation of the density evolution is given in (15).

$$m_u(l) = \sum_j^{d_r} \rho_j \phi^{-1} \left(1 - \left[\int_{-\infty}^{\infty} \sum_i^{d_l} \lambda_i \tanh\left(\frac{x}{2}\right) P_{v,i} \times (x, x_s, m_u(l-1)) dx \right]^{j-1} \right) \quad (15)$$

where $P_{v,i}(x, x_s, m_u(l-1)) = (P(\mathcal{L}, x_s) * N((i-1)m_u(l-1), 2(i-1)m_u(l-1)))(x)$, $P_{v,i}(\mathcal{L}, x_s, m_u(0)) = P(\mathcal{L}, x_s) \forall i$,

and $P(\mathcal{L}, x_s) = \int_0^{\infty} \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{hx_s}\right) \exp\left(-\frac{\left(\mathcal{L} - \frac{(hx_s)^2}{2\sigma^2}\right)^2}{2\frac{(hx_s)^2}{\sigma^2}}\right) f_H(h) dh$.

The BER is given as

$$P_B = \sum_i^{d_l} \lambda_i \int_{-\infty}^0 P_{v,i}(x, x_s, m_u(l)) dx. \quad (16)$$

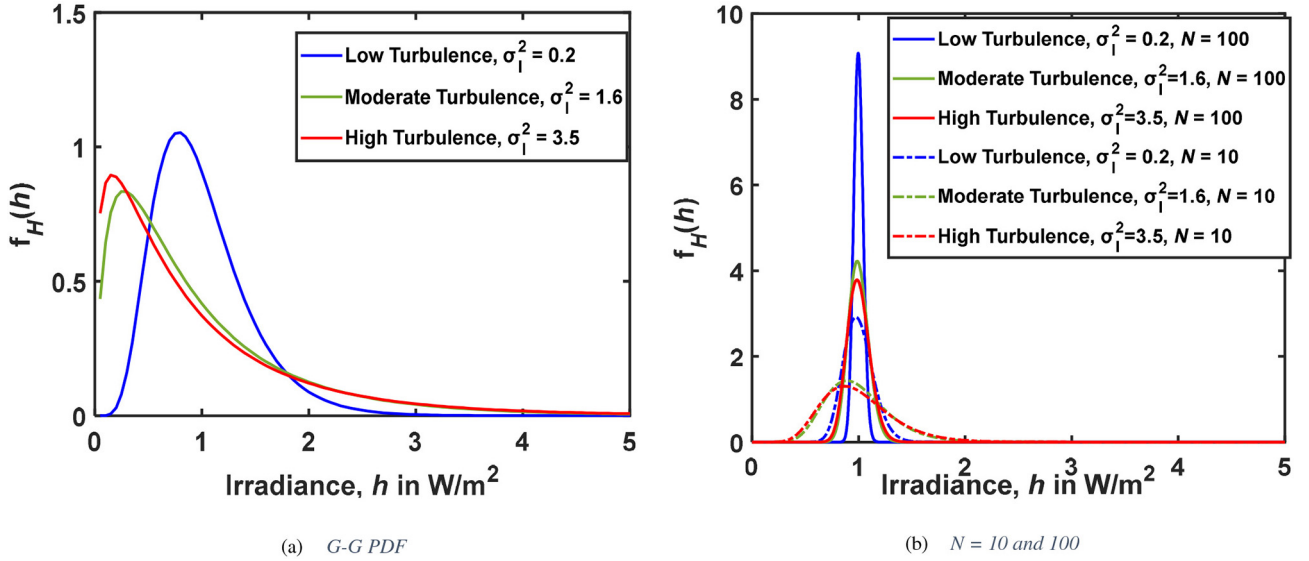


FIGURE 3. PDF for G-G and fast fading channel.

V. PERFORMANCE IN A FAST FADING CHANNEL

We implement a fast fading channel by considering multiple channel paths through which signal transmits, and hence t_c is much smaller than t_b [13]. We then model the SIMO/MIMO channel to compare the performance of a fast fading channel. In this section, we evaluate the capacity and BER performance of a fast fading channel and spatially multiplexed channel under different turbulence regimes with LDPC codes.

A. FAST FADING CHANNEL

We first derive the channel pdf in a fast fading channel, considering N channel coefficients being independent and G-G distributed. We simulate the channel pdf for a value of $N = 10$ and $N = 100$ [10], i.e., we assume that each transmitted bits experience N different channel coefficients through which data gets transmitted. The chosen values of N cover a large landscape of fast fading paradigm. The channel pdf is modeled as

$$f_{\tilde{H}}(h) = (f_{H_1} * f_{H_2} * \dots * f_{H_N})(h), \quad (17)$$

where $*$ is the convolution operation, $f_{H_1}, \dots, f_{H_N}$ denote the G-G channel pdf. The resulting pdf is obtained as

$$f_H(h) = N \times f_{\tilde{H}}(Nh). \quad (18)$$

The simulations are performed using the above channel model. We plot the G-G pdf and pdf of the fast fading channel for two different values of N , i.e., 10 and 100 (see, (18)) in Fig. 3 (a) and Fig. 3 (b), respectively. It can be observed from Fig. 3 (b) that as N increases, the pdf squishes due to the reduced variance ($\lim_{N \rightarrow \infty} \frac{\sigma^2}{N}$). Note that as $N \rightarrow \infty$, the performance approaches that of an AWGN channel.

1) BELIEF PROPAGATION ALGORITHM IN A FAST FADING CHANNEL

In a fast fading channel, we consider one coded bit sees N different and independent channel coefficients which result in, $t_c < t_b$. The received signal is processed to evaluate the bit reliabilities (LLRs). We compute the LLRs as described in Section IV-A1, where h is modeled using the fast fading channel pdf (see, (18)). Assuming that all-zero codeword is transmitted, we get the pdf of LLR in the fast fading channel as given in (19).

We integrate (19) to get a pdf of LLR in a fast fading channel. We show $P(\mathcal{L}_f, x_s, N)$ in Fig. 4 (a) for $N = 10$. The other characteristics mean, normalized distance defined as $d(\Lambda) = \frac{m(\Lambda)}{\sigma(\Lambda)}$ and the probability of error of the channel LLR ($\int_{-\infty}^0 P(\mathcal{L}_f, x_s, N) d\mathcal{L}_f$) as a function of SNR (dB) are given in Fig. 4 (b), (c), and (d), respectively. The pdf of LLR obtained for the fast fading channel in all the three turbulence regimes almost overlap because the channel effects are averaged out for each information bit in a fast fading channel. Further, in a fast fading channel, the mean of the LLR increases as the SNR increases. Also, the normalized distance rises slowly. We observe that the probability of channel error decreases with the increase in SNR for all the turbulence regimes.

$$P(\mathcal{L}_f, x_s, N) = \int_0^\infty \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{hx_s} \right) \exp \left(-\frac{\left(\mathcal{L}_f - \frac{(hx_s)^2}{2\sigma^2} \right)^2}{2 \frac{(hx_s)^2}{\sigma^2}} \right) N \times f_{\tilde{H}}(Nh) dh. \quad (19)$$

The pdf of v_i for the fast fading channel in the l th iteration is given as

$$P_{v,i}(x, x_s, N, m_u(l-1)) = (P(\mathcal{L}_f, x_s, N) * N((i-1)m_u(l-1), 2(i-1)m_u(l-1)))(x), \quad (20)$$

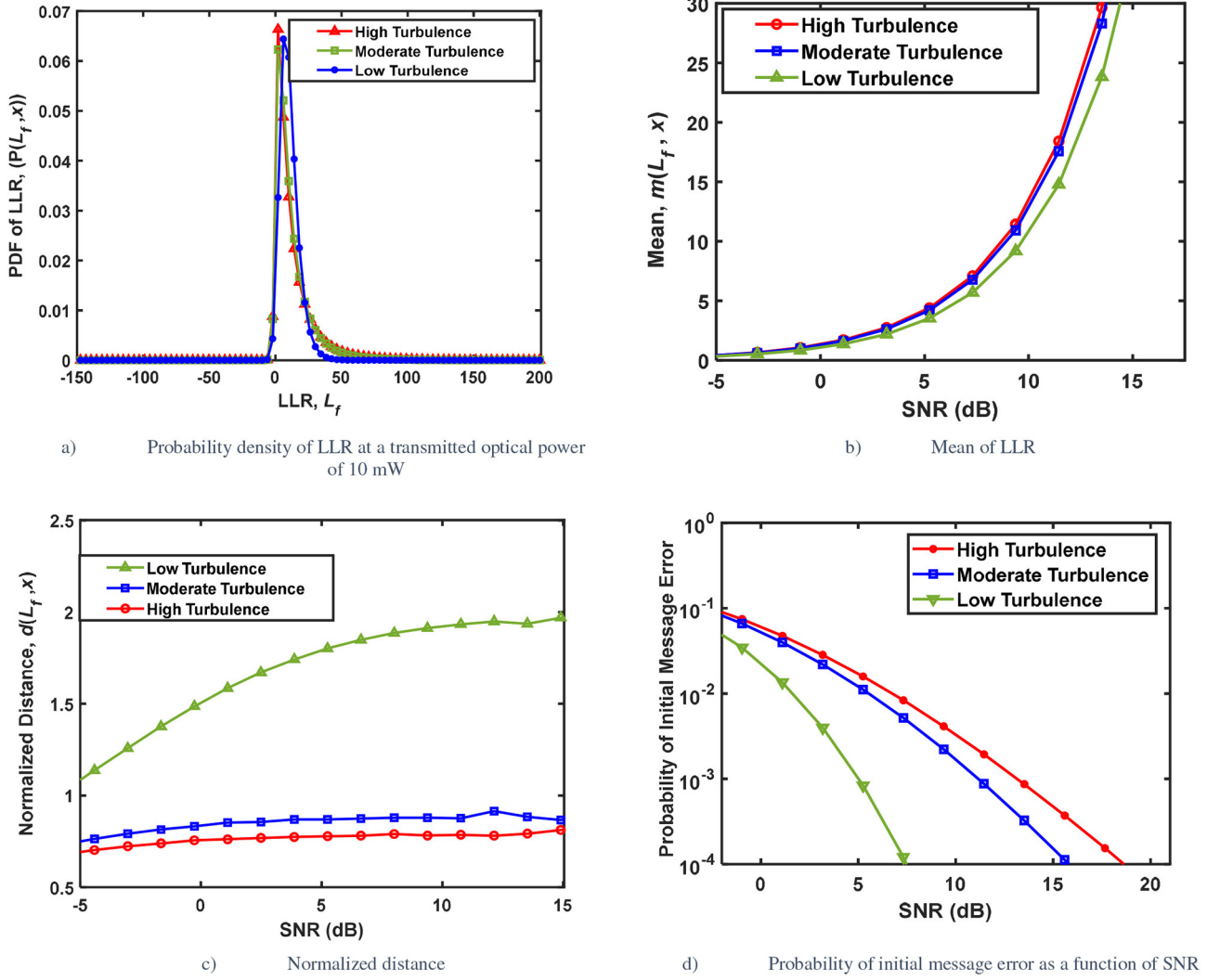


FIGURE 4. Characteristics of the LLR for a fast fading channel.

The final equation of the density evolution is

$$m_u(l) = \sum_j \rho_j \phi^{-1} \left(1 - \left[\int_{-\infty}^{\infty} \int_i^{d_l} \lambda_i \int_{-\infty}^{\infty} \tanh\left(\frac{x}{2}\right) P_{v,i} \times (x_s, N, m_u(l-1)) dx \right]^{j-1} \right), \quad (21)$$

Further, the BER is given as

$$P_B(x_s, N, m) = \sum_i \lambda_i \int_{-\infty}^0 P_{v,i}(x, x_s, N, m_u(l-1)) dx. \quad (22)$$

B. SPATIAL MULTIPLEXING

To compare the performance of the fast fading channel with $N = 10$, we use spatial multiplexing for an LDPC coded FSO channel. We use the SIMO technique of 1×10 configuration and MIMO with 2×5 configuration to make the analysis alike. We show the schematic for SIMO/MIMO configuration

in Fig. 5. The LDPC encoded data is fed to the data repetition block for MIMO configuration. The received signal for the k_{th} receiver is given as

$$Y_k = X \sum_{m=1}^M H_{k,m} + I_{th,k}, \quad (23)$$

where $I_{th,k}$ is the thermal noise at the k_{th} receiver and $H_{k,m}$ the intensity fading vector between the m_{th} optical source (either 1 or 2) and a k_{th} optical receiver (either 10 or 5). The received data is combined using an equal gain combiner (EGC) at the receiver, where the signal from individual transmission path is scaled with equal weight ($w_i = 1$, $i \in \{1, k\}$). The transmitter section shown in Fig. 5 with the gray color represents MIMO configuration. In MIMO configuration, we ensure that the total transmit power is the same as that of a system with no transmit diversity. Similarly, K receiver aperture areas are kept the same as the aperture area of a system with no receive diversity.

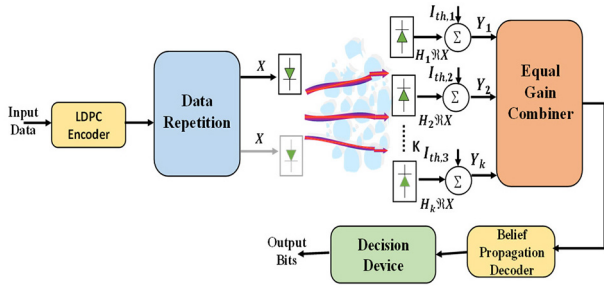


FIGURE 5. Schematic of SIMO and MIMO LDPC coded FSO system.

1) BP ALGORITHM FOR A SIMO/MIMO CHANNEL

In SIMO/MIMO configuration, we compute the LLRs, assuming that each fading path is spatially uncorrelated. It implies that the distance between the neighboring receivers is greater than fading correlation length. The LLR for SIMO configuration for the j th coded bit with EGC combiner is given as

$$\mathcal{L}_{SIMO,j} = \sum_{k=1}^K \left(\frac{2x_j y_j h_j[k] - x_j^2 h_j^2[k]}{2\sigma^2} \right), \quad (24)$$

and for MIMO configuration for j th coded bit is given as

$$\mathcal{L}_{MIMO,j} = \sum_{k=1}^K \left(\frac{2x_j y_j \sum_{m=1}^M h_j[k, m] - x_j^2 \sum_{m=1}^M h_j^2[k, m]}{2\sigma^2} \right). \quad (25)$$

The LLRs for every receiver are fed as an input to the BP decoder. The pdf of LLR for SIMO configuration is given in (26)

$$P(\mathcal{L}_{SIMO,j}, x_j) = \int_0^\infty \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{hx_j} \right) \exp \left(- \frac{\left(\mathcal{L}_{SIMO,j} - \frac{\sum_{k=1}^K (h_j[k]x_j)^2}{2\sigma^2} \right)^2}{2 \frac{\sum_{k=1}^K h_j^2[k]x_j^2}{\sigma^2}} \right) \times f_{H,SIMO}(h) dh. \quad (26)$$

$$P(\mathcal{L}_{MIMO,j}, x_j) = \int_0^\infty \frac{1}{\sqrt{2\pi}} \left(\frac{\sigma}{hx_j} \right) \exp \left(- \frac{\left(\mathcal{L}_{MIMO,j} - \frac{\sum_{k=1}^K \sum_{m=1}^M (h_j[k, m]x_j)^2}{2\sigma^2} \right)^2}{2 \frac{\sum_{k=1}^K \sum_{m=1}^M h_j^2[k, m]x_j^2}{\sigma^2}} \right) \times f_{H,MIMO}(h) dh, \quad (27)$$

In SIMO configuration, multiple diverse paths result in averaging of the scintillations over the multiple channel paths. This reduces the irradiance fluctuations by a factor of K and SNR required to achieve a target BER reduces by a factor of $\sqrt{K}$ [31] when we compare with no diversity configuration. Thus, the scintillation parameters α and β of the channel pdf gets modified as $\alpha_K = \frac{\alpha(1+K\beta)}{\beta+1}$ and $\beta_K = K\beta$ [31]. The pdf $f_{H,SIMO}(h)$ is modified by plugging

TABLE 4. Optimized degree distribution pair for fast fading channel/spatially multiplexed channel.

Turbulence Regime	$\rho(x)$	$\lambda(x)$
Weak	$0.0002x^2 + 0.9998x^3$	$0.9959x + 0.0041x^2$
Moderate	$0.3x^5 + 0.7x^6$	$0.5201x + 0.1997x^2 + 0.2802x^3$
Strong	$0.3x^5 + 0.7x^6$	$0.4684x + 0.2692x^2 + 0.2624x^3$

these parameters in (2). The rest of the analysis remains same as given in Section IV-A1. Similarly, for the MIMO configuration, pdf of the initial LLR is given in (27). In MIMO configuration, with multiple diverse paths, the scintillation parameters $\alpha_S = S\alpha + \varepsilon$ and $\beta_S = S\beta$ where $S = MK$ and the error term, ε is given as [32]

$$\varepsilon = (S - 1) \frac{-0.127 - 0.95\alpha - 0.0058\beta}{1 + 0.00124\alpha + 0.98\beta}. \quad (28)$$

We show PDF of LLR, i.e., $P(\mathcal{L}_{SIMO/MIMO}, x)$ obtained for a diversity condition in Fig. 6 (a). We integrate (26) and (27) for all values of h to get a pdf of LLR for a SIMO/MIMO FSO channel. We once again plot other characteristics viz., mean, normalized distance, and the probability of error of the channel as a function of SNR (dB) in Fig. 6 (b), (c), and (d), respectively.

2) RESULTS AND DISCUSSIONS

This subsection discusses the performance of regular and irregular LDPC codes under a fast fading and spatially multiplexed channel. We use the regular (3,6) LDPC codes, and the irregular LDPC codes degree distribution pairs designed at the desired code rate [9] of 0.5 and are reported in Table 4. We present the simulation and analytical results of BER vs. SNR for the LDPC coded system in a fast fading ($N = 10$) and spatially multiplexed system in Figs. 7–9. The performance is compared to the uncoded OOK system for the corresponding channel condition. The performance gain is reported in Table 5.

We consider the same system parameters for both the systems, but since in a fast fading channel, $1/N^{th}$ bit goes through a different channel fades, averaging the channel for one bit makes the BER performance poorer than in a spatially multiplexed channel where each bit goes through an independent fading path. These results can be validated from the capacity analysis carried out in Section III (see, Tables 2 and 3). We observe that spatial diversity (1×10) gives the best BER performance when compared with the MIMO (2×5) configuration or the fast fading channel condition. This is because in diversity configurations (SIMO), each bit goes through an independent channel path, and at the receiver, data received from each independent link are combined using EGC. Multiple paths facilitate the reception of data more reliably than a SISO or a MIMO link. Hence, we

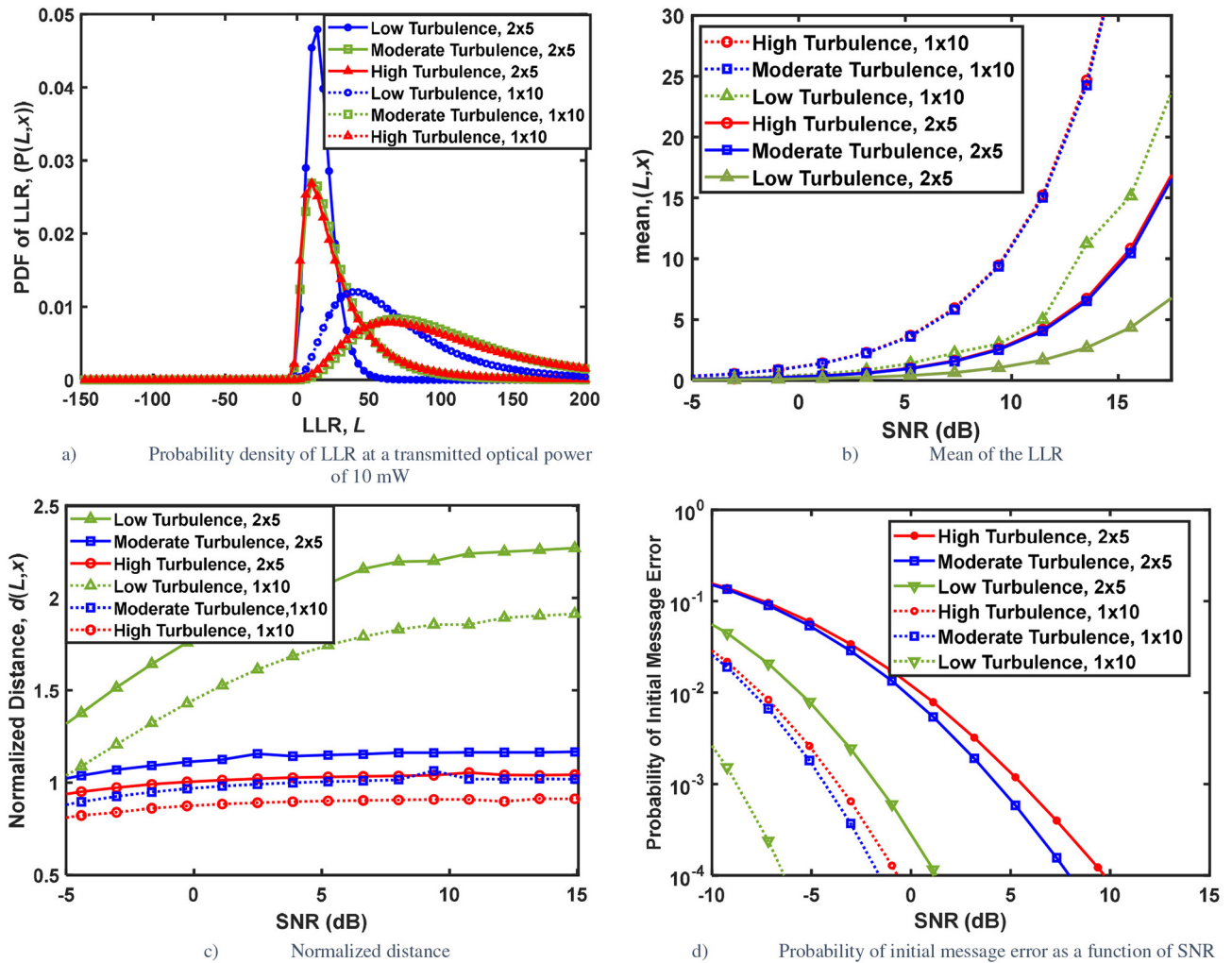


FIGURE 6. Characteristics of the LLR for a spatial multiplexed channel.

TABLE 5. Performance in fast fading and spatial multiplexed channel.

Schemes	Coding Gain (dB)			Gap To Capacity (dB)		
	High Turbulence	Moderate Turbulence	Low Turbulence	High Turbulence	Moderate Turbulence	Low Turbulence
Irregular LDPC, $N = 10$	13.95	12.79	9.76	12.20	10.60	9.32
Regular LDPC, $N = 10$	12.00	10.84	8.46	13.25	13.00	11.90
Irregular LDPC, 1×10	17.83	11.40	8.39	18.83	18.22	17.76
Regular LDPC, 1×10	16.02	9.97	7.30	19.80	19.01	18.67
Irregular LDPC, 2×5	13.19	17.36	8.43	23.70	22.59	22.45
Regular LDPC, 2×5	12.17	15.52	7.32	25.35	23.86	23.20

obtain better performance with LDPC coded diversity system than LDPC code with SISO or MIMO configuration.

We also report the gap to capacity in Table 5 (Capacity ref. Table 3). It can be observed from Table 5 that a maximum

coding gain of 17.83 dB is obtained from the 1x10 MIMO system. The coding gain at a target BER is defined as the difference in dB between the SNR required to achieve the target BER with the coded system (LDPC coded in our case)

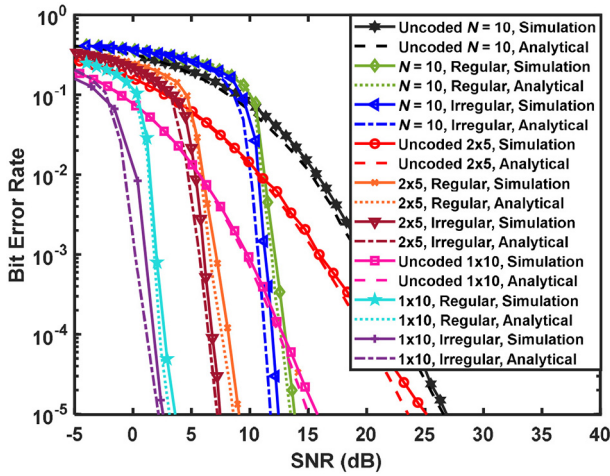


FIGURE 7. Analytical and simulation results for $\sigma_f^2 = 3.5$ (high).

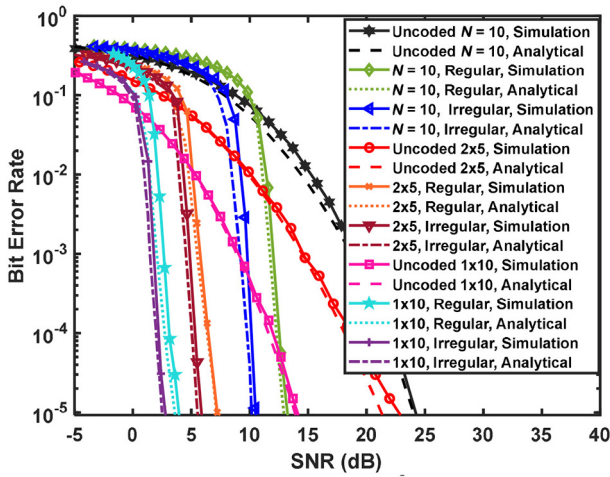


FIGURE 8. Analytical and simulation results for $\sigma_f^2 = 1.6$ (moderate).

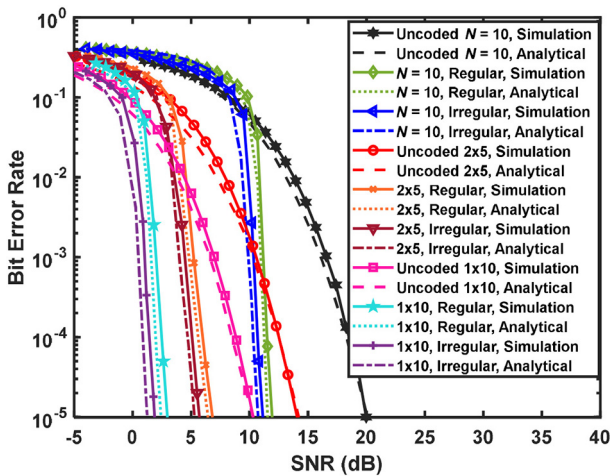


FIGURE 9. Analytical and simulation results with interleaver for $\sigma_f^2 = 0.2$ (low).

and the SNR required to achieve the target BER with an uncoded system. We incur a higher capacity gap in spatial diversity and MIMO systems when we compare with the

fast fading channel. Also, the irregular LDPC codes in all the systems under consideration achieve higher coding gain and lesser gap to capacity than the corresponding regular LDPC codes.

VI. CONCLUSION

In this paper, we evaluate the performance of LDPC codes in an FSO communication link under various channel conditions – slow fading, fast fading, and spatially multiplexed channel with low, moderate, and high levels of turbulence. For the evaluation, we propose analytical models that are verified through simulations. We analyze and formulate the BP algorithm of LDPC codes in a fast fading and spatially multiplexed FSO channel. We enumerate the coding gains and gap with Shannon's limit (capacity) to visualize the difference we need to cross over to achieve the capacity. We obtain the maximum coding gain of 13.95 dB and 17.83 dB and minimum gap to the capacity of 9.32 dB and 17.76 dB in a fast fading channel and spatial multiplexing case, respectively. Though this work is proposed for slow fading FSO channels, it can also be used for other slow fading channels, e.g., visible light communication.

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